

Projective varieties with almost nef tangent sheaves and its dynamical application

Workshop on Algebraic Geometry over complex number field or in
positive characteristic

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1. Background & Motivation

- Unless otherwise stated, we work over the field $k = \bar{k}$ of char $k = 0$.
- The **tangent sheaf** T_X of a normal projective variety X is defined as the dual of its sheaf of Kähler differentials Ω_X .
- Let D be an effective reduced divisor on a normal variety X , and U a big open subset over which (X, D) is log smooth. Let $\Omega_X^{[1]}(\log D)$ be the reflexive sheaf on X whose restriction to U is the sheaf of the usual logarithmic differential 1-form. Its dual $T_X(-\log D)$ is called the **logarithmic tangent sheaf**.
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- **For example**, a normal toric pair has trivial logarithmic tangent sheaf.
- A surjective endomorphism $f: X \rightarrow X$ of a projective variety X is said to be **int-amplified**, if $f^*H - H$ is ample for some ample divisor H
 $\iff$ the induced linear operation $f^*|_{N^1(X)}$ on the Néron-Severi space has all of the modulus > 1 (**Meng, 2020**).
- **For example**, any non-isomorphic surjective endomorphism of a Fano manifold of Picard number 1 has to be int-amplified.

1. Background & Motivation

Motivation

One of the central topics in algebraic geometry is to classify varieties.

- (1) from the aspect of positivity of tangent sheaves;
- (2) from the aspect of algebraic dynamics;
- Both of them share some similar features on the geometry of the underlying space: **two fundamental building blocks** and **equivariant minimal model program**;
- Varieties admitting non-isomorphic endomorphisms are expected to be good candidates for concrete examples admitting good positivity of tangent sheaves (associated with $\text{Aut}^0(X)$);
- Re-establish the structure theorems for varieties with certain positive tangent sheaves via running a suitable minimal model program.

dynamics $\implies$ positivity $\implies$ geometry.

1. Background & Motivation

We recall the following positivity of reflexive sheaves.

Definition (Positivity of reflexive sheaves)

Let $\mathcal{E}$ be a torsion free coherent sheaf on a normal projective variety X .

- $\mathcal{E}$ is **pseudo-effective** (psef) or **weakly positive** if $\forall a \in \mathbb{Z}_{>0}$ and $\forall$ ample divisor A , $\exists b \in \mathbb{Z}_{>0}$ such that $\text{Sym}^{[ab]} \mathcal{E} \otimes \mathcal{O}_X(bA)$ is globally generated at a general point (Viehweg, 1983), (Nakayama, 2004), (Matsumura, 2023); this is stronger than $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(1)$ being psef.
- $\mathcal{E}$ is **ample** (resp. **nef**) if the tautological line bundle $\mathcal{O}_{\mathbb{P}_X(\mathcal{E})}(1)$ is ample (resp. nef) on the Grothendieck projectivization $\mathbb{P}_X(\mathcal{E})$.
- $\mathcal{E}$ is **almost nef** if the base field is **uncountable** and $\exists$ countably many proper subvarieties $Z_i \subseteq X$ such that the sheaf $\mathcal{E}|_C$ is nef, $\forall$ curve $C \not\subseteq \bigcup_i Z_i$ (Demailly-Peternell-Schneider, 2001).
- $\mathcal{E}$ is **numerically flat** if both $\mathcal{E}$ and its dual $\mathcal{E}^\vee$ are nef and locally free.

1. Background & Motivation (Building blocks)

(A) For both positivity of tangent sheaves and algebraic dynamics, there are two fundamental building blocks of algebraic varieties:

Theorem (Mori, 1979)=Hartshorne conjecture; cf. (Siu-Yau, 1980)

Every irreducible n -dimensional non-singular projective variety with ample tangent bundle defined over a field $k = \bar{k}$ of char $k = 0$ is isomorphic to the projective space $\mathbb{P}_k^n$.

Theorem (Demailly-Peternell-Schneider, 1994)

A compact Kähler manifold with nef tangent bundle, up to a finite étale cover, admits a locally trivial fibration over a complex torus whose fibre is a Fano manifold with nef tangent bundle.

Very recently, (Wang, 2024) proved that the tangent bundle of such a fibre is big, partially confirming a conjecture of Campana-Peternell.

1. Background & Motivation (Building blocks)

(A) For both positivity of tangent sheaves and algebraic dynamics, there are two fundamental building blocks of algebraic varieties:

Theorem (Hosono-Iwai-Matsumura, 2022), (Iwai, 2022)

A complex projective manifold with psef (resp. almost nef) tangent bundle, up to a finite étale cover, admits a smooth fibration to an abelian variety whose fibre is rationally connected and has psef (resp. almost nef) tangent bundle.

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Theorem (Meng-Zhang, 2018), (Meng, 2020), (Yoshikawa, 2021)

Let X be a $\mathbb{Q}$ -factorial projective klt variety admitting an int-amplified endomorphism. Then there exists a quasi-étale finite cover $\tilde{X} \rightarrow X$ such that the Albanese morphism $\text{alb}_{\tilde{X}}$ is a fibre space and of Fano type over an abelian variety.

In fact, when X is smooth, it is not hard to prove that $\text{alb}_{\tilde{X}}$ is smooth.

1. Background & Motivation (MMP)

(B1) For varieties with psef tangent sheaves, they behave well along the MMP:

Theorem (Matsumura, 2023)

Let X be a complex projective $\mathbb{Q}$ -factorial klt variety with psef tangent sheaf. Then there exist finitely many projective varieties $\{X_k\}_{k=0}^N$ and $\{X'_k\}_{k=0}^N$ with

$$X = X_0 \dashrightarrow X'_0 \rightarrow X_1 \dashrightarrow X'_1 \rightarrow \cdots \rightarrow X_{N-1} \dashrightarrow X'_{N-1} \rightarrow X_N$$

satisfying the following conditions:

- X_k and X'_k are klt with psef tangent sheaf;
- each $X_k \dashrightarrow X'_k$ is birational obtained from the composite of divisorial contractions and flips;
- each $X'_k \rightarrow X_{k+1}$ is a Mori fibre space; and
- the end product X_N is either a point or a $\mathbb{Q}$ -abelian variety.

1. Background & Motivation (MMP)

(B2) For certain algebraic dynamics, they also behave well along the MMP:

Theorem (Meng-Zhang, 2018), (Meng, 2020)

Let f be an int-amplified endomorphism of a projective $\mathbb{Q}$ -factorial klt variety X . Then, there are either a point or a Q -abelian variety Y , and an f -equivariant relative MMP over Y (i.e., $f^s = f_1$ for some s descends to f_i on each X_i)

$$X = X_1 \dashrightarrow \cdots \dashrightarrow X_i \dashrightarrow \cdots \dashrightarrow X_r = Y$$

with $X_i \dashrightarrow X_{i+1}$ a divisorial contraction, a flip, or a Mori fibre space (and $X_{r-1} \rightarrow X_r$ a Mori fibre space) such that

- If K_X is psef, then $X = Y$ is Q -abelian.
- If K_X is not psef, then for each i , f_i is int-amplified and $X_i \rightarrow Y$ is an equi-dimensional morphism with every fibre irreducible.
- All fibres are rationally connected if the base field is uncountable.

2. Structure theorem I: almost nef tangent sheaves

In the joint work with Iwai and Matsumura, we generalize the structure theorems for projective manifolds with certain positive tangent bundles to the case of klt varieties.

Theorem (Iwai-Matsumura-Z., 23)

Let X be a complex projective klt variety with almost nef tangent sheaf. Then, there is a fibration $\alpha: X \rightarrow Y$ satisfying the following properties:

- (1) The fibration α is flat, and every irreducible component of the singular locus of X (if exists) dominates Y .
- (2) The base variety Y is an étale quotient of an abelian variety.
- (3) Any fiber of α is an irreducible and reduced rationally connected klt variety, and a very general fiber has almost nef tangent sheaf.
- (4) If moreover, the canonical divisor K_X is psef, then X is smooth (as an étale quotient of an abelian variety).

We expect α to be a locally trivial fibration.

2. Structure theorem I: almost nef tangent sheaves

As a key step to prove the structure theorem, we need the characterization of flat sheaves, extending (Cao-Höring, 2019), (Höring-Peternell, 2019), and (Hosono-Iwai-Matsumura, 2022).

Proposition (Iwai-Matsumura-Z., 2023)

Let X be a complex projective klt variety of dimension n , and $\mathcal{E}$ a psef or almost nef reflexive sheaf on X . If $\det(\mathcal{E}) \cdot H^{n-1} = 0$ for an ample divisor H , then there is a quasi-étale cover such that the reflexive pullback of $\mathcal{E}$ is numerically flat (locally free).

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- By certain transcendental argument in (Meng-Z., 2024), the above proposition holds for arbitrary $k = \bar{k}$ of char $k = 0$ (for almost nef, we need an additional assumption that the base field k is uncountable).
- The above theorem plays a significant role in the next section on the dynamically invariant reflexive sheaves.

2. Structure theorem II: flat logarithmic tangent sheaves

Theorem (Druel-Lo Bianco, 2022); cf. (Ejiri-Yoshikawa, 2023)

Let X be a normal projective variety of klt type, and $D \geq 0$ a reduced divisor on X such that (X, D) is log canonical. Suppose that $\Omega_X^{[1]}(\log D)$ is locally free $\mathbb{R}$ -flat, and $-(K_X + D)$ is nef. Then $\exists$ a smooth projective variety Q with $K_Q \equiv 0$ and a surjective morphism with connected fibres $a : X \rightarrow Q$. The fibration $(X, D) \rightarrow Q$ is locally trivial for the analytic topology and any fibre F is a toric variety with boundary divisor $D|_F$. Moreover, Q contains no rational curves.

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- The condition "R-flat" is weaker than numerical flatness.
- If we assume the logarithmic differential sheaf is even numerically flat, then from the generically surjective map $T_X(-\log D) \rightarrow T_X$, we know that T_X is almost nef and thus the variety Q has to be an étale quotient of an abelian variety by (Hosono-Iwai-Matsumura, 22), or (Iwai, 2022) or (Iwai-Matsumura-Z., 2023).

2. Structure theorem II: flat logarithmic tangent sheaves

Use certain transcendental arguments again, we can prove the following.

Theorem (Meng-Z., 2024); essentially due to (Druel-Lo Bianco, 22)

Let X be a projective klt variety over $k = \bar{k}$ of char $k = 0$. Let $D \geq 0$ be a reduced divisor such that $\Omega_X^{[1]}(\log D)$ is numerically flat and locally free. Then the augmented irregularity $\tilde{q}(X) < \infty$ holds. Suppose further that (X, D) is log canonical with $q(X) = \tilde{q}(X)$. Then the albanese morphism $(X, D) \rightarrow A$ is an analytically locally trivial toric fibration.

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- Note that the albanese morphism is independent of the choice of base field; however, it is not clear for the MRC fibration.
- To use the transcendental argument, one needs to be careful for the extension from the results on MRC fibration over $\mathbb{C}$ to the results on MRC fibration over countable field (e.g., certain K3 surfaces over $\overline{\mathbb{Q}}$).
- Under the dynamical assumption, we can see later that one can even make the toric fibration Zariski locally trivial.

3. Toric characterization: totally invariant ramifications

Setup (totally invariant ramifications – TIR)

Let X be a projective klt variety admitting an int-amplified endomorphism f and D a reduced divisor such that $f^{-1}(D) = D$ and f is quasi-étale away from D , that is $f^*(K_X + D) = K_X + D$.

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Several motivations to study the geometry of TIR.

- (Hwang-Nakayama, 2011) observed that when X is a Fano manifold of Picard number 1 such that either (1) X is almost homogeneous or (2) the VMRT of X along a general point is a finite union of linear spaces, any surjective endomorphism satisfies the case TIR.
- (Meng-Zhang, 2022) and (Matsuzawa-Meng-Shibata-Zhang, 2022) applied the equivariant minimal model program and reduced certain arithmetic conjectures for the case of int-amplified endomorphisms to the case TIR.

3. Toric characterization: totally invariant ramifications

An observation (Meng-Z., 2024)

Under the setup of TIR, the logarithmic differential sheaf $\Omega_X^{[1]}(\log D)$ is numerically flat up to a finite quasi-étale cover.

More generally, we prove the following:

Proposition (Meng-Z., 2024)

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X of dimension n . Let $\mathcal{E}$ be a reflexive sheaf on X such that the reflexive pullback $f^{[*]}\mathcal{E} \cong \mathcal{E}$. Then $\mathcal{E}$ is psef. Moreover, $\pi^{[*]}\mathcal{E}$ is numerically flat locally free for some quasi-étale cover $\pi: Y \rightarrow X$ if X is further klt.

When $\mathcal{E}$ is locally free, the above proposition follows immediately from the dynamical assumption as well as the Simpson's correspondence by showing that both $c_1(\mathcal{E})$ and $c_2(\mathcal{E})$ are weakly numerically trivial.

3. Toric characterization: totally invariant ramifications

By (Druel-Lo Bianco, 2022), a technical analysis of the boundary divisor, and the dynamical lifting, we obtain the following structure theorem.

Theorem (Meng-Z., 2024)

Under the setup of TIR, there exists an f -equivariant quasi-étale cover $\pi: \widehat{X} \rightarrow X$ such that the pair $(\widehat{X}, \pi^*D)$ admits a Zariski locally trivial toric fibration over an abelian variety A . Moreover, there exists an f -equivariant generically finite cover $\varphi: \widetilde{X} \rightarrow \widehat{X} \rightarrow X$ with the additional property that the pair $(\widetilde{X}, \varphi^*D)$ is SNC with $\widetilde{X}$ being smooth.

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Under the assumption of TIR:

- When X is a Fano manifold of Picard number 1, (Hwang-Nakayama, 2011) proved that X itself is isomorphic to the projective space.
- When X is a smooth rationally connected variety, (Meng-Zhang, 2019) and (Meng-Z., 2022) proved that X is a toric variety.
- When X is a Fano type variety: (Moraga-Yanez-Yeong, 2024) proved that up to a finite cover, X is toric (a completely different approach).

3. Toric characterization: totally invariant ramifications

- The locally trivial toric fibration $(\widehat{X}, \pi^*D) \rightarrow A$ being Zariski locally trivial is equivalent to the restriction of each irreducible component of D to a fibre remaining irreducible.
- As an application of our structure theorem, together with previous works ([Meng-Zhang, 2022](#)) and ([Matsuzawa-Meng-Shibata-Zhang, 2023](#)), we can complete the proof of certain arithmetic theorems for the case of int-amplified endomorphisms (not introduced herein).

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- As an application of our structure theorem, together with previous works (Meng-Zhang, 2022) and (Matsuzawa-Meng-Shibata-Zhang, 2023), we can complete the proof of certain arithmetic theorems for the case of int-amplified endomorphisms (not introduced herein).
- We conjectured the following:

Conjecture (Meng-Z., 2024); cf. (Yoshikawa, 2021)

Let X be a smooth projective variety admitting an int-amplified endomorphism. Then up to replacing X by an étale cover, the Albanese map is a Zariski locally trivial toric fibration onto an abelian variety.

The hardest part is on the rationally connected building block: ✓ surfaces (Nakayama, 2002) and Fano 3-folds (Amerik-Rovinsky-Van de Ven, 1999), (Hwang-Mok, 2003), (Meng-Zhang-Z., 2022)

3. Toric characterization: totally invariant ramifications

We briefly explain the rough idea of the proof of the following proposition.

Proposition (Meng-Z., 2024)

Let $f: X \rightarrow X$ be an int-amplified endomorphism of a normal projective variety X of dimension n . Let $\mathcal{E}$ be a reflexive sheaf on X such that the reflexive pullback $f^{[*]}\mathcal{E} \cong \mathcal{E}$. Then $\mathcal{E}$ is psef. Moreover, $\pi^{[*]}\mathcal{E}$ is numerically flat locally free for some quasi-étale cover $\pi: Y \rightarrow X$ if X is further klt.

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- Since f is int-amplified, $f^*H - H$ is ample for some ample divisor H .
- With H replaced by a sufficiently large multiple, we can show that $\mathrm{Sym}^{[m]}\mathcal{E} \otimes \mathcal{O}_X(H)$ is psef for any $m \in \mathbb{Z}_{>0}$.
- We follow the spirit of an argument of (Ejiri-Iwai-Matsumura, 2023) or (Matsumura, 2023) showing that the pseudo-effectiveness of a reflexive sheaf $\mathcal{F}$ is equivalent to the pseudo-effectiveness of its reflexive pullback $f^{[*]}\mathcal{F}$.

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- We can then prove that $\mathcal{E}$ itself is psef (thinking it as the "limit" of $\mathrm{Sym}^{[m]}\mathcal{E} \otimes \mathcal{O}_X(H)$).
- Taking the determinant, we have $f^{[*]}(c_1(\mathcal{E})) = c_1(\mathcal{E})$.
- As any int-amplified endomorphism has all the eigenvalues of modulus > 1 for the linear operation f^* on the cycle classes (Meng, 2020), we obtain that $c_1(\mathcal{E}) \cdot H^{n-1} = 0$.
- We conclude the numerical flatness by (Iwai-Matsumura-Z., 2023).

4. Further Questions

Implicitly, the dynamical assumption should impose certain positivity of the tangent bundle.

Question

Let X be a rationally connected (or even Fano) smooth projective variety admitting an int-amplified endomorphism. Does the tangent bundle T_X satisfy certain positivity?

- Projective toric varieties have trivial logarithmic tangent sheaves and hence have psef and almost nef tangent sheaves.
- Smooth projective toric varieties have (weakly) big tangent bundles (Hsiao, 2015).

Question

Can we establish the equivariant minimal model program for projective klt varieties with other positivity of tangent sheaves? (e.g. almost nef?)

Thank you for your attention!